Novel Method: Electrochemical Selective Recovery

Strategic Metal Extraction from Electronic Waste (E-Waste)

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1. Introduction

Electronic waste contains valuable strategic metals such as copper, silver, gold, palladium and other elements. Conventional recovery techniques may require high temperatures or strong chemical reagents. **Electrochemical Selective Recovery** provides an alternative approach in which electrical energy is used to separate and recover metals from a metal-rich solution.

2. Proposed Novel Method

In this method, e-waste is first mechanically processed to expose and concentrate metal-containing components. The material is then treated to produce a solution containing dissolved metal ions. This solution is placed in an electrochemical cell containing an anode and a cathode. When a controlled electrical potential is applied, metal ions undergo reduction and are deposited as solid metals on the cathode.

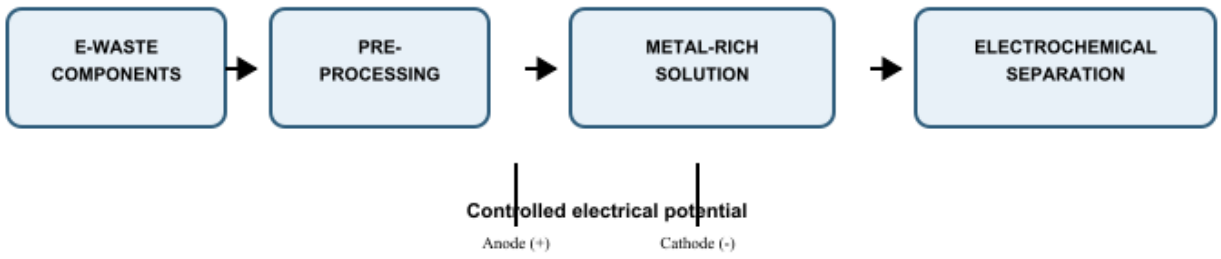

Figure 1: Overall process flow of electrochemical selective recovery.

3. Working Principle

The method works on the principle of oxidation-reduction reactions. Metal ions present in the solution move under the influence of an electric field. By controlling the voltage, current and operating conditions, metals can be selectively deposited on the cathode. The recovered deposit can then be collected, purified and processed for reuse.

4. Electrochemical Recovery Cell

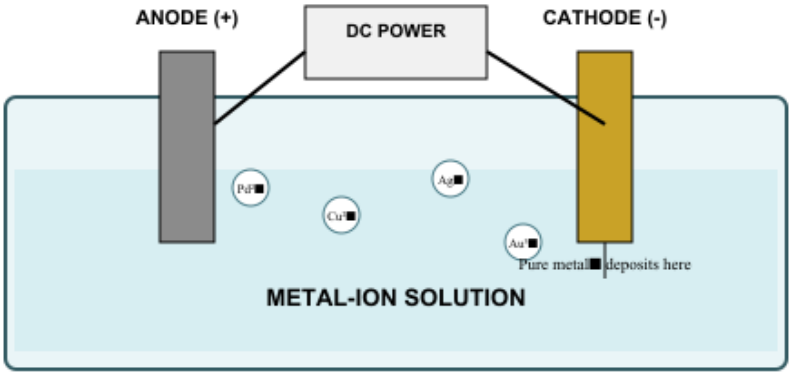

Figure 2: Schematic representation of an electrochemical recovery cell.

5. Step-by-Step Process

Step 1: Collection and Pre-processing

Electronic waste is collected, dismantled and mechanically processed to separate metal-containing components from non-metallic materials.

Step 2: Preparation of Metal-Rich Solution

The processed material is treated so that valuable metals are converted into dissolved metal ions in a solution.

Step 3: Electrochemical Separation

The metal-rich solution is placed in an electrochemical cell. A controlled electrical potential is applied between the electrodes.

Step 4: Selective Deposition

Metal ions are reduced and deposited as solid metals on the cathode. The operating conditions can be adjusted for selective recovery.

Step 5: Collection and Reuse

The deposited metals are removed, purified if required and sent for reuse in new electronic products or industrial applications.

6. Key Parameters

Parameter	Role in the Process
Applied potential / voltage	Controls the electrochemical reaction and deposition of metal ions.
Current density	Influences the rate and quality of metal deposition.
Electrolyte composition	Affects metal-ion movement and selective recovery.
Electrode material	Influences efficiency, stability and purity of recovered metal.

7. Novelty and Advantages

The novelty of this method lies in the use of controlled electrical energy to selectively separate strategic metals from e-waste. Unlike methods based primarily on thermal treatment or aggressive chemical processing, electrochemical recovery allows the process to be precisely controlled according to the electrochemical properties of different metals.

Advantage	Description
Selective recovery	Electrical conditions can be adjusted to target specific metal ions.
High-purity deposition	Recovered metals can be deposited directly as solid metal layers.
Reduced chemical dependence	The process can reduce the need for large quantities of harsh chemical reagents.
Process control	Voltage, current and operating conditions can be precisely monitored.
Resource recovery	Valuable metals from e-waste can be returned to the manufacturing cycle.

8. Applications

• Recovery of copper from printed circuit boards.

• Recovery of silver, gold and palladium from electronic components.

• Treatment of metal-rich solutions generated during e-waste recycling.

• Recovery and purification of strategic metals for reuse in electronics and other industries.

9. Conclusion

Electrochemical Selective Recovery is a promising novel method for sustainable metal extraction from electronic waste. By applying controlled electrical energy, valuable metal ions can be selectively converted into recoverable solid metals. This approach supports the circular economy by reducing the need for new raw-material extraction and enabling valuable resources to be recovered from discarded electronic products.

Proposed Process Flow

E-Waste Components → **Pre-processing** → **Metal-Rich Solution** → **Electrochemical Separation** → **Selective Metal Deposition** → **Pure Strategic Metals**